Tighter security for XMSS (Reduction-Based Approach)

Work in Progress

Abstract

Contents

1 Introduction

We attempt the reproduce the proof strategy of [1] [1], with two main changes:

1. Keep using Multi-Target Collision Resistance (SM-TCR) and Multi-Target Collision Resistance with Random Sampling (SM-rTCR), but replace Multi-Target Preimage Resistance (SM-PRE) and Multi-Target Undetectability (SM-UD) with a new (non-standard) property: Multi-Target Chain Preimage Resistance (SM-CHAIN).

2. Multi-user scenario.

The idea behind 1. is to avoid the $L \cdot v \cdot 2^w$ factor in the final security bound. This overhead comes from the event in which an adversary is able to find a direct preimage in a "hash chain" at a position just before the "earlier" revealed value. In [1], the reduction must guess which chain and which position the adversary will target, leading to the multiplicative loss. By capturing this attack scenario directly in SM-CHAIN, we can obtain a tighter reduction that avoids this guessing overhead. It remains to prove a bound for SM-CHAIN in the ROM (doable) and QROM (not trivial).

1.1 Tweakable Hash Functions

Definition 1.1 (Tweakable Hash Function)**.** Consider sets $\mathcal{H}$ (the hash space), $\mathcal{P}$ (the public parameters space), $\mathcal{T}$ (the tweak space), and let $\mathcal{M}$ (the message space). A tweakable hash function is an efficiently computable function

$$\mathsf{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}.$$

1.2 Public Parameter Collisions

Definition 1.2. We denote by $\mathcal{E}_{\text{distinct PP}}$ the event: "all public parameters PP across the N signers are distinct".

Proposition 1.3. $\Pr[\neg\mathcal{E}_{distinct\ PP}] \leq \frac{N^2}{2^{\mathsf{pp}+1}}$

Proof. Each signer samples $\mathsf{PP} \in \{0,1\}^{\mathsf{pp}}$ independently and uniformly at random. By A.1 with $k = N$ values sampled from a set of size 2^{pp}, the probability of any collision is at most $N^2/(2 \cdot 2^{\mathsf{pp}}) = N^2/2^{\mathsf{pp}+1}$. $\square$

Assumption: We condition the rest of our analysis on $\mathcal{E}_{\text{distinct PP}}$ (note that $\mathcal{E}_{\text{distinct PP}}$ is independent from the adversary), i.e. we will focus on bounding:

$$\mathsf{Adv}^{\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}}_{\mathsf{SIG},N distinct}(\mathcal{A}) := \Pr[\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}_{\mathsf{SIG},N}(\mathcal{A}) \Rightarrow 1 \mid \mathcal{E}_{\text{distinct PP}}].$$

As a result, all the properties defined below will assume that the public parameters of different users are distinct.

1.3 Multi-user properties

1.3.1 Multi-User Multi-Target Collision Resistance

Definition 1.4 (Multi-User Multi-Target Collision Resistance)**.** Let $\mathsf{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}$ be a tweakable hash function. Let $N \in \mathbb{N}$ be the number of users, let $p \in \mathbb{N}$ be a per-user query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\mathsf{MULTI\text{-}SM\text{-}TCR}_{\mathsf{Th},N,p}(\mathcal{A})$:

Setup:

[1] Many parts of the current document are a direct reproduction of [1]

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $Q_i := \emptyset$.

Query Phase: Run $\mathcal{A}$ with (classical) access to the following oracle:

- Hash Oracle $\mathcal{O}_{\mathsf{target}}(i, T, M)$ where $i \in [N]$, $T \in \mathcal{T}$, $M \in \mathcal{M}$:
  (a) If $|Q_i| \geq p$ or there exists $M' \in \mathcal{M}$ with $(T, M') \in Q_i$, return $\perp$.
  (b) Insert (T, M) into Q_i.
  (c) Return $\mathsf{Th}(P_i, T, M)$.

Reveal Phase: When $\mathcal{A}$ signals to proceed:

1. Disable $\mathcal{O}_{\mathsf{target}}$ and provide $(P_1, \dots, P_N)$ to $\mathcal{A}$.

Output: $\mathcal{A}$ outputs (i^*, j^*, M^*) with $i^* \in [N]$, $j^* \in [|Q_{i^*}|]$, and $M^* \in \mathcal{M}$.

Winning Condition: Let (T_{j^*}, M_{j^*}) be the j^*th entry in Q_{i^*}. Output 1 if and only if:

(i) $\mathsf{Th}(P_{i^*}, T_{j^*}, M_{j^*}) = \mathsf{Th}(P_{i^*}, T_{j^*}, M^*)$, and

(ii) $M^* \neq M_{j^*}$.

The advantage is:

$$\mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}TCR}}_{\mathsf{Th},N,p}(\mathcal{A}) := \Pr[\mathsf{MULTI\text{-}SM\text{-}TCR}_{\mathsf{Th},N,p}(\mathcal{A}) \Rightarrow 1].$$

Intuition 1.5. An attacker plays a game where they first *commit* to a collection of messages—without knowing the hash function's secret parameters—and then later attempt to find a *different* message that collides with one of their committed messages.

1.3.2 Multi-User Multi-Target Collision Resistance with Random Sampling

Definition 1.6 (Multi-User Multi-Target Collision Resistance with Random Sampling)**.** Let $K \in \mathbb{N}$ be an integer. Let $\mathsf{Th} : \mathcal{P} \times \mathcal{T} \times (\mathcal{M} \times \mathcal{R}) \to \mathcal{H}$ be a tweakable hash function, where the input is split into a message part $M \in \mathcal{M}$ and a randomness part $\rho \in \mathcal{R}$. Let $\mathsf{Prop} : \mathcal{H} \to \{0, 1\}$ be a predicate on the output space. Let $N \in \mathbb{N}$ be the number of users, let $p \in \mathbb{N}$ be a per-user query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\mathsf{MULTI\text{-}SM\text{-}rTCR}^{K}_{\mathsf{Th},N,p,\mathsf{Prop}}(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $Q_i := \emptyset$.

Query Phase: Run $\mathcal{A}$ with input $(P_1, \dots, P_N)$ and (classical) access to the following oracle:

- Sampling Oracle $\mathcal{O}_{\mathsf{sample}}(i, T, M)$ where $i \in [N]$, $T \in \mathcal{T}$, $M \in \mathcal{M}$:
  (a) If $|Q_i| \geq p$ or there exists a tuple $(T, M', \rho') \in Q_i$ for some M', ρ', return $\perp$.
  (b) Set $\mathsf{ctr} := 0$ and $x := \perp$. While $\mathsf{ctr} < K$ and $x = \perp$:
    i. Sample $\rho \xleftarrow{\$} \mathcal{R}$.

ii. Set $x := \mathsf{Th}(P_i, T, M, \rho)$.
iii. If $\mathsf{Prop}(x) = 1$: Insert (T, M, ρ) into Q_i.
iv. Else: Set $x := \perp$, $\rho := \perp$.
v. Set $\mathsf{ctr} := \mathsf{ctr} + 1$.

(c) If $x = \perp$: Insert $(T, M, \perp)$ into Q_i.
(d) Return (x, ρ).

Output: $\mathcal{A}$ outputs (i^*, j^*, M^*, ρ^*) with $i^* \in [N]$, $j^* \in [|Q_{i^*}|]$, $M^* \in \mathcal{M}$, and $\rho^* \in \mathcal{R}$.

Winning Condition: Let $(T_{j^*}, M_{j^*}, \rho_{j^*})$ be the j^*th entry in Q_{i^*}. Output 1 if and only if:

(i) $\mathsf{Th}(P_{i^*}, T_{j^*}, M_{j^*}, \rho_{j^*}) = \mathsf{Th}(P_{i^*}, T_{j^*}, M^*, \rho^*)$, and

(ii) $(M^*, \rho^*) \neq (M_{j^*}, \rho_{j^*})$.

The advantage is:

$$\mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}rTCR},K}_{\mathsf{Th},N,p,\mathsf{Prop}}(\mathcal{A}) := \Pr[\mathsf{MULTI\text{-}SM\text{-}rTCR}^{K}_{\mathsf{Th},N,p,\mathsf{Prop}}(\mathcal{A}) \Rightarrow 1].$$

1.3.3 Multi-User Multi-Target Chain Preimage Resistance

Definition 1.7 (Multi-User Multi-Target Chain Preimage Resistance)**.** Let $\mathsf{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \to \mathcal{H}$ be a tweakable hash function with $\mathcal{H} \subseteq \mathcal{M}$. Let $w \geq 1$ be the chain length parameter, let $N \in \mathbb{N}$ be the number of users, let $p \in \mathbb{N}$ be a per-user query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\mathsf{MULTI\text{-}SM\text{-}CHAIN}_{\mathsf{Th},N,w,p}(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $Q_i := \emptyset$, $\mathsf{Chains}_i := []$, and counters $\mathsf{ctr}_i := 0$.

Query Phase: Run $\mathcal{A}$ with (classical) access to the Chain Oracle $\mathcal{O}_{\mathsf{chain}}(i, T_1, \ldots, T_{2^w-1})$ for $i \in [N]$, $(T_1, \ldots, T_{2^w-1}) \in \mathcal{T}^{2^w-1}$:

(a) If $\mathsf{ctr}_i \geq p$, return $\perp$.

(b) If $|\{T_1, \ldots, T_{2^w-1}\}| < 2^w - 1$ or $\{T_1, \ldots, T_{2^w-1}\} \cap Q_i \neq \emptyset$, return $\perp$.

(c) Sample $Y_0 \xleftarrow{\$} \mathcal{H}$.

(d) Compute $Y_j := \mathsf{Th}(P_i, T_j, Y_{j-1})$ for $j = 1, \ldots, 2^w - 1$.

(e) Append $\big((T_1, \ldots, T_{2^w-1}), (Y_0, \ldots, Y_{2^w-1})\big)$ to Chains_i.

(f) Set $Q_i := Q_i \cup \{T_1, \ldots, T_{2^w-1}\}$ and $\mathsf{ctr}_i := \mathsf{ctr}_i + 1$.

(g) Return Y_{2^w-1}.

Reveal Phase: When $\mathcal{A}$ signals to proceed:

1. Disable $\mathcal{O}_{\mathsf{chain}}$ and provide $(P_1, \ldots, P_N)$ to $\mathcal{A}$.
2. For each $i \in [N]$ and $j \in [|\mathsf{Chains}_i|]$, initialize $\mathsf{Revealed}_{i,j} := \emptyset$.
3. Give $\mathcal{A}$ (classical) access to the Reveal Oracle $\mathcal{O}_{\mathsf{reveal}}(i, j, k)$ for $i \in [N]$, $j \in [|\mathsf{Chains}_i|]$, $k \in \{0, \ldots, 2^w - 2\}$:

(a) Let $(Y_0^{i,j}, \ldots, Y_{2^w-1}^{i,j})$ be the chain values for user i, chain j.
(b) Set $\mathsf{Revealed}_{i,j} := \mathsf{Revealed}_{i,j} \cup \{k\}$.
(c) Return $Y_k^{i,j}$.

Output: $\mathcal{A}$ outputs (i^*, j^*, x) with $i^* \in [N]$, $j^* \in [|\mathsf{Chains}_{i^*}|]$, and $x \in \mathcal{H}$.

Winning Condition: Let $\big((T_1^{i^*,j^*}, \ldots, T_{2^w-1}^{i^*,j^*}), (Y_0^{i^*,j^*}, \ldots, Y_{2^w-1}^{i^*,j^*})\big)$ be the j^*th entry in Chains_{i^*}. Let $k^* := \min(\mathsf{Revealed}_{i^*,j^*} \cup \{2^w - 1\})$ be the earliest revealed position (or $2^w - 1$ if nothing was revealed). Output 1 if and only if $k^* \neq 0$ and:

$$\mathsf{Th}(P_{i^*}, T_{k^*}^{i^*,j^*}, x) = Y_{k^*}^{i^*,j^*}.$$

The advantage is:

$$\mathsf{Adv}_{\mathsf{Th},N,w,p}^{\mathsf{MULTI\text{-}SM\text{-}CHAIN}}(\mathcal{A}) := \Pr[\mathsf{MULTI\text{-}SM\text{-}CHAIN}_{\mathsf{Th},N,w,p}(\mathcal{A}) \Rightarrow 1].$$

Intuition 1.8. The challenger constructs secret hash chains of the form:

$$Y_0 \xrightarrow{T_1} Y_1 \xrightarrow{T_2} Y_2 \xrightarrow{T_3} \cdots \xrightarrow{T_{2^w-1}} Y_{2^w-1}$$

but initially only reveals the *final* value Y_{2^w-1}. Later, the attacker may query to see some intermediate values. The attacker wins by finding a preimage of the earliest revealed value in the chain.

1.3.4 Multi-User Synchronized Security

Definition 1.9 (Multi-User Synchronized Security)**.** Let $\mathsf{SIG} = (\mathsf{Gen}, \mathsf{Sig}, \mathsf{Ver})$ be a synchronized signature scheme with lifetime L, let $N \in \mathbb{N}$ be the number of users, and let $\mathcal{A}$ be any algorithm. Consider the following experiment $\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}_{\mathsf{SIG},N}(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, generate keys $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{Gen}(\mathsf{par})$.
2. For each $i \in [N]$, initialize $\mathsf{Signed}_i[\cdot] := \perp$ (a map indexed by epochs).

Query Phase: Run $\mathcal{A}$ on input par and $(\mathsf{pk}_1, \ldots, \mathsf{pk}_N)$, with (classical) access to the following oracle:

- Signing Oracle $\mathcal{O}_{\mathsf{Sig}}(i, \mathsf{ep}, m)$ where $i \in [N]$, $\mathsf{ep} \in [L]$, $m \in \{0,1\}^{\ell_{\mathsf{msg}}}$:
  (a) If $\mathsf{Signed}_i[\mathsf{ep}] \neq \perp$, return $\perp$.
  (b) Compute $\sigma \leftarrow \mathsf{Sig}(\mathsf{sk}_i, \mathsf{ep}, m)$.
  (c) Set $\mathsf{Signed}_i[\mathsf{ep}] := (m, \sigma)$.
  (d) Return σ.

Output: $\mathcal{A}$ outputs a forgery $(i^*, \mathsf{ep}^*, m^*, \sigma^*)$ with $i^* \in [N]$, $\mathsf{ep}^* \in [L]$, and $m^* \in \{0,1\}^{\ell_{\mathsf{msg}}}$.

Winning Condition: Output 1 if and only if:

(i) $\mathsf{Ver}(\mathsf{pk}_{i^*}, \mathsf{ep}^*, m^*, \sigma^*) = 1$, and
(ii) $(m^*, \sigma^*) \neq \mathsf{Signed}_{i^*}[\mathsf{ep}^*]$.

The advantage is:

$$\mathsf{Adv}_{\mathsf{SIG},N}^{\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}}(\mathcal{A}) := \Pr[\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}_{\mathsf{SIG},N}(\mathcal{A}) \Rightarrow 1].$$

Remark 1.10 (Strong Unforgeability). This multi-user definition captures *strong unforgeability*: a forgery is valid even if it is for a message that was previously signed, as long as the signature itself is different.

1.3.5 Multi-User Target Collision Resistance for Incomparable Encodings

Definition 1.11 (Multi-User Target Collision Resistance for IE)**.** Let $\mathsf{IncEnc} : \mathcal{P} \times \{0,1\}^{\ell_{\mathsf{msg}}} \times \mathcal{R} \times [L] \to \mathcal{C} \cup \{\perp\}$ be an incomparable encoding scheme with code $\mathcal{C} \subseteq \{0, \ldots, 2^w - 1\}^v$. Let $K \in \mathbb{N}$ be an integer, let $N \in \mathbb{N}$ be the number of users, and let $p \in \mathbb{N}$ be a per-user query bound. For any algorithm $\mathcal{A}$, consider the following experiment $\mathsf{MULTI\text{-}T\text{-}COLL\text{-}RES}^K_{\mathsf{IncEnc},N,p}(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $L_i := \emptyset$.

Query Phase: Run $\mathcal{A}$ with input $(P_1, \ldots, P_N)$ and (classical) access to the following oracle:

- Encoding Oracle $\mathcal{O}_{\mathsf{enc}}(i, m, \mathsf{ep})$ where $i \in [N]$, $m \in \{0,1\}^{\ell_{\mathsf{msg}}}$, $\mathsf{ep} \in [L]$:
  (a) If there exists an entry $(m', \rho', \mathsf{ep}, x') \in L_i$ (with the same ep) or $|L_i| \geq p$, return $\perp$.
  (b) Set $\mathsf{ctr} := 0$ and $x := \perp$. While $\mathsf{ctr} < K$ and $x = \perp$:
    i. Sample $\rho \xleftarrow{\$} \mathcal{R}$.
    ii. Set $x := \mathsf{IncEnc}(P_i, m, \rho, \mathsf{ep})$.
    iii. Set $\mathsf{ctr} := \mathsf{ctr} + 1$.
  (c) If $x = \perp$: Insert $(m, \perp, \mathsf{ep}, \perp)$ into L_i and return $\perp$.
  (d) Else (note: $x \in \mathcal{C}$): Insert $(m, \rho, \mathsf{ep}, x)$ into L_i and return (x, ρ).

Output: $\mathcal{A}$ outputs $(i^*, m^*, \rho^*, \mathsf{ep}^*)$ with $i^* \in [N]$, $m^* \in \{0,1\}^{\ell_{\mathsf{msg}}}$, $\rho^* \in \mathcal{R}$, and $\mathsf{ep}^* \in [L]$.

Winning Condition: Compute the encoding $x^* := \mathsf{IncEnc}(P_{i^*}, m^*, \rho^*, \mathsf{ep}^*)$. Output 1 if and only if:

(i) $x^* \neq \perp$, and

(ii) there exists a pair $(m, \rho) \neq (m^*, \rho^*)$ with $(m, \rho, \mathsf{ep}^*, x^*) \in L_{i^*}$.

The advantage is:

$$\mathsf{Adv}^{\mathsf{MULTI\text{-}T\text{-}COLL\text{-}RES},K}_{\mathsf{IncEnc},N,p}(\mathcal{A}) := \Pr[\mathsf{MULTI\text{-}T\text{-}COLL\text{-}RES}^K_{\mathsf{IncEnc},N,p}(\mathcal{A}) \Rightarrow 1].$$

1.4 Heuristic Bounds in the (Quantum) Random Oracle Model

We reuse the same bounds from [1] for SM-TCR and SM-rTCR, scaled by a factor of N to account for the multi-user setting (trivial reduction). We could derive a tighter bound for SM-rTCR (without the N factor) but this would not improve the final result for XMSS significantly, so we avoid this additional effort.

The bounds for SM-CHAIN remain to be proven. Classical bound: Soon. Quantum bound: Harder.

Notion	Classical Bound	Quantum Bound
$\mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}TCR}}_{\mathsf{Th},N,p}(\mathcal{A})$	$N \cdot \left(\frac{2q+1}{\lvert\mathcal{H}\rvert} + \frac{2q}{\lvert\mathcal{P}\rvert}\right)$	$N \cdot \left(\frac{32(q+1)^2}{\lvert\mathcal{H}\rvert} + \frac{32q^2}{\lvert\mathcal{P}\rvert}\right)$
$\mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}rTCR},K}_{\mathsf{Th},N,p,\mathsf{Prop}}(\mathcal{A})$	$N \cdot \left(\frac{q'+1}{\lvert\mathcal{H}\rvert} + \frac{q' \cdot pK}{\lvert\mathcal{R}\rvert}\right)$	$N \cdot \left(\frac{8(q'+1)^2}{\lvert\mathcal{H}\rvert} + \frac{3pK}{2} \cdot \sqrt{\frac{q'}{\lvert\mathcal{R}\rvert}}\right)$
$\mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}CHAIN}}_{\mathsf{Th},N,w,p}(\mathcal{A})$	$\frac{q+1}{\lvert\mathcal{H}\rvert}$ (?)	$\frac{O(q^2)}{\lvert\mathcal{H}\rvert}$ (?)

Table 1: Upper bounds on the success probability of an adversary against properties of tweakable hash functions $\mathsf{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \to \mathcal{H}$, when the hash function is modeled as a (classical or quantum) random oracle. For SM-rTCR, we assume $\mathcal{M} = \mathcal{M}_0 \times \mathcal{R}$. Here, q is the number of (quantum or classical) queries to the hash function and p is the number of classical queries to the challenge oracle, K denotes the number of queries the challenge oracle in SM-rTCR makes to the hash function. We set $q' := q + pK$.

2 Security of Generalized XMSS

Theorem 2.1 (Multi-User Security of Generalized XMSS)**.** *Consider the scheme* $\mathsf{SIG}[\mathsf{IncEnc}, \mathsf{Th}, K]$ *with parameters* n, v, w, L, $K \in \mathbb{N}$ *and* Th, IncEnc *as in Construction 3. Let* $N \in \mathbb{N}$ *be the number of users. Then, conditioned on the event in which all signer's public parameter are distinct (which happens with probability* $1 - \Pr[PP\ collision] \geq 1 - N^2/2^{\mathsf{pp}+1}$*, independently from the adversary), every algorithm* $\mathcal{A}$*, that makes no more than* q_s *signature queries (total across all users), there are algorithms* $\mathcal{B}_i$ *with* $T(\mathcal{A}) \approx T(\mathcal{B}_i)$ *for all* i *and*

$$\mathsf{Adv}^{\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}}_{\mathsf{SIG}[\mathsf{IncEnc},\mathsf{Th}],N distinct}(\mathcal{A})$$

$$\leq 2 \cdot \mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}TCR}}_{\mathsf{Th},N,2\cdot L\cdot v\cdot 2^w}(\mathcal{B}_1) + \mathsf{Adv}^{\mathsf{MULTI\text{-}T\text{-}COLL\text{-}RES},K}_{\mathsf{IncEnc},N,q_s}(\mathcal{B}_2) + \mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}CHAIN}}_{\mathsf{Th},N,w,L\cdot v}(\mathcal{B}_4).$$

Proof. Write $\mathsf{SIG} := \mathsf{SIG}[\mathsf{IncEnc}, \mathsf{Th}, K]$. We prove the statement by giving a sequence of games. For the ith game, denoted by $\mathsf{Game}.i$, we let $\mathsf{Adv}^{\mathsf{Game}.i}_{\mathsf{SIG},N}(\mathcal{A})$ be the probability that the game outputs 1.

Game.0: Our starting point is the multi-user synchronized security game for adversary $\mathcal{A}$ and scheme SIG with N users, see Definition 1.9. To recall, the game first generates N key pairs $(\mathsf{pk}_i, \mathsf{sk}_i)$ for $i \in [N]$ as in the scheme and then gives the public keys $\mathsf{pk}_1, \ldots, \mathsf{pk}_N$ to $\mathcal{A}$, where $\mathsf{pk}_i = (\mathsf{root}_i, P_i)$. The adversary then gets access to an oracle $\mathcal{O}_{\mathsf{Sig}}(i, \mathsf{ep}, m)$ to obtain signatures for messages m and epochs ep from user $i \in [N]$. The oracle can only be called once per user-epoch pair, and stores the resulting message-signature pair (m, σ) as $\mathsf{Signed}_i[\mathsf{ep}] := (m, \sigma)$. Finally, the adversary outputs a forgery $(i^*, \mathsf{ep}^*, m^*, \sigma^*)$ and wins if $\mathsf{Ver}(\mathsf{pk}_{i^*}, \mathsf{ep}^*, m^*, \sigma^*) = 1$ and $(m^*, \sigma^*) \neq \mathsf{Signed}_{i^*}[\mathsf{ep}^*]$. In the scheme we consider, signatures have the form $\sigma = (\rho, \sigma_{\mathsf{OTS}}, \mathsf{path}_{\mathsf{ep}})$. That is, the second part of the winning condition states that $(m^*, (\rho^*, \sigma^*_{\mathsf{OTS}}, \mathsf{path}^*_{\mathsf{ep}^*})) \neq \mathsf{Signed}_{i^*}[\mathsf{ep}^*]$, where $\sigma^* = (\rho^*, \sigma^*_{\mathsf{OTS}}, \mathsf{path}^*_{\mathsf{ep}^*})$. We also make the following assumption, which is without loss of generality: we have $\mathsf{Signed}_{i^*}[\mathsf{ep}^*] \neq \perp$ at the end of the game, i.e., $\mathcal{A}$ queried the signing oracle for user i^* at the forgery epoch. By definition, we have

$$\mathsf{Adv}^{\mathsf{MULTI\text{-}SY\text{-}UF\text{-}CMA}}_{\mathsf{SIG}[\mathsf{IncEnc},\mathsf{Th}],N}(\mathcal{A}) = \mathsf{Adv}^{\mathsf{Game}.0}_{\mathsf{SIG},N}(\mathcal{A}).$$

Game.1: We now change the winning condition. Informally, we rule out that the adversary forges by breaking the security of the Merkle tree for any user. More precisely, for each user $i \in [N]$, denote the list of one-time public keys that the game generated during key generation by

$\mathsf{pk}_{i,1}, \dots, \mathsf{pk}_{i,L}$, i.e., $\mathsf{root}_i = \mathsf{Root}(P_i, \mathsf{pk}_{i,1}, \dots, \mathsf{pk}_{i,L})$. Further, let $\sigma^* = (\rho^*, \sigma^*_{\mathsf{OTS}}, \mathsf{path}^*_{\mathsf{ep}})$ be $\mathcal{A}$'s forgery for user i^*, and let $\mathsf{pk}^*_{i^*,\mathsf{ep}^*}$ denote the one-time public key for this user-epoch that the verification algorithm recomputes from σ^*_{OTS} and $x^* = \mathsf{IncEnc}(P_{i^*}, m^*, \rho^*, \mathsf{ep}^*)$. Let $\mathsf{path}_{i^*,\mathsf{ep}^*} := \mathsf{Path}(P_{i^*}, \mathsf{pk}_{i^*,1}, \dots, \mathsf{pk}_{i^*,L}, \mathsf{ep}^*)$ be the Merkle path that the signing oracle $\mathcal{O}_{\mathsf{Sig}}(i^*, \mathsf{ep}^*, \cdot)$ would include in signatures. Now, the game $\mathsf{Game.1}$ would output 0 if

$$(\mathsf{pk}_{i^*,\mathsf{ep}^*}, \mathsf{path}_{i^*,\mathsf{ep}^*}) \neq (\mathsf{pk}^*_{i^*,\mathsf{ep}^*}, \mathsf{path}^*_{\mathsf{ep}^*}). \tag{1}$$

Here, the left hand side is what honest signing would compute for user i^*, and the right hand side is derived from the forgery. Otherwise, the game checks the winning condition as before. The games only differ if Equation (1) holds. We will now argue that this event can be bounded by a reduction $\mathcal{B}_1$ breaking multi-user multi-target collision resistance, i.e.,

$$\left|\mathsf{Adv}^{\mathsf{Game.0}}_{\mathsf{SIG},N}(\mathcal{A}) - \mathsf{Adv}^{\mathsf{Game.1}}_{\mathsf{SIG},N}(\mathcal{A})\right| \leq \mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}TCR}}_{\mathsf{Th},N,2\cdot L\cdot v\cdot 2^w}(\mathcal{B}_1).$$

To understand how such a reduction $\mathcal{B}_1$ works, consider the part of the Merkle tree for user i^* that is revealed when opening the leaf at position ep^*. Assume $(\mathsf{pk}_{i^*,\mathsf{ep}^*}, \mathsf{path}_{i^*,\mathsf{ep}^*}) \neq (\mathsf{pk}^*_{i^*,\mathsf{ep}^*}, \mathsf{path}^*_{\mathsf{ep}^*})$. As in the single-user case, there must be a collision somewhere in the Merkle tree for user i^*.

In the reduction, we use the multi-user multi-target collision resistance game (Definition 1.4). In the query phase, for each user $i \in [N]$, we first generate secret elements $\mathsf{sk}_{i,j,k}$ for $j \in [L]$, $k \in [v]$ uniformly at random. Next, we query $\mathcal{O}_{\mathsf{target}}(i, \cdot, \cdot)$ with the corresponding secret values and tweaks to build all chains and compute $\mathsf{pk}_{i,\mathsf{ep}}$ for all epochs $\mathsf{ep} \in [L]$. We then build the Merkle tree for user i using the oracle $\mathcal{O}_{\mathsf{target}}(i, \cdot, \cdot)$. In this way, we can set up all chains and all Merkle trees without explicit access to the parameters $P_1, \dots, P_N$. After the reveal phase, we obtain $P_1, \dots, P_N$ from the game. We use (root_i, P_i) as the public key for user i and give all public keys to $\mathcal{A}$. As we know all secret keys, we can perfectly simulate the rest of $\mathsf{Game.0}$ for $\mathcal{A}$. If for the forgery it holds that $(\mathsf{pk}_{i^*,\mathsf{ep}^*}, \mathsf{path}_{i^*,\mathsf{ep}^*}) \neq (\mathsf{pk}^*_{i^*,\mathsf{ep}^*}, \mathsf{path}^*_{\mathsf{ep}^*})$, there must be a collision in the Merkle tree for user i^* as explained above. By recomputing the root of the Merkle tree from the forged signature we find this collision efficiently, thereby breaking multi-user multi-target collision resistance.

Game.2: This is the same as $\mathsf{Game.1}$ but we let the game output 0 if we can extract a collision for the incomparable encoding scheme. More precisely, recall that we consider only the case in which $\mathcal{O}_{\mathsf{Sig}}(i^*, \mathsf{ep}^*, \cdot)$ has been queried, and let $(m, \sigma) = \mathsf{Signed}_{i^*}[\mathsf{ep}^*]$, where $\sigma = (\rho, \sigma_{\mathsf{OTS}}, \mathsf{path}_{\mathsf{ep}^*})$. We let the game output 0 if we have

$$(m, \rho) \neq (m^*, \rho^*) \land \mathsf{IncEnc}(P_{i^*}, m, \rho, \mathsf{ep}^*) = \mathsf{IncEnc}(P_{i^*}, m^*, \rho^*, \mathsf{ep}^*). \tag{2}$$

Otherwise, the game outputs what $\mathsf{Game.1}$ would output. We bound the difference between these two games using a reduction $\mathcal{B}_2$ against multi-user target collision resistance of IncEnc (Definition 1.11). The reduction works as follows:

1. Receive $(P_1, \dots, P_N)$ as input from the $\mathsf{MULTI\text{-}T\text{-}COLL\text{-}RES}^K_{\mathsf{IncEnc},N,q_s}$ game, together with access to the encoding oracle $\mathcal{O}_{\mathsf{enc}}$.

2. For each user $i \in [N]$, generate secret key components: sample $\mathsf{sk}_{i,\mathsf{ep},j} \xleftarrow{\$} \mathcal{H}$ for all $\mathsf{ep} \in [L]$, $j \in [v]$, and compute the corresponding public key components $\mathsf{pk}_{i,\mathsf{ep},j}$ by chaining, then build the Merkle tree to obtain root_i. Set $\mathsf{pk}_i := (\mathsf{root}_i, P_i)$.

3. Run $\mathcal{A}$ on input $(\mathsf{pk}_1, \dots, \mathsf{pk}_N)$ and simulate the signing oracle $\mathcal{O}_{\mathsf{Sig}}(i, \mathsf{ep}, m)$ as follows:
   - Query $\mathcal{O}_{\mathsf{enc}}(i, m, \mathsf{ep})$ to obtain (x, ρ) where $x = \mathsf{IncEnc}(P_i, m, \rho, \mathsf{ep})$.
   - If $x = \perp$, return $\perp$.

- Compute the OTS signature σ_{OTS} using the secret keys $\mathsf{sk}_{i,\mathsf{ep},1}, \ldots, \mathsf{sk}_{i,\mathsf{ep},v}$ and the encoding x.
- Compute the Merkle path $\mathsf{path}_{\mathsf{ep}}$ and return $\sigma := (\rho, \sigma_{\mathsf{OTS}}, \mathsf{path}_{\mathsf{ep}})$.

4. When $\mathcal{A}$ outputs a forgery $(i^*, \mathsf{ep}^*, m^*, \sigma^*)$ where $\sigma^* = (\rho^*, \sigma^*_{\mathsf{OTS}}, \mathsf{path}^*_{\mathsf{ep}^*})$:
   - If Equation (2) holds, output $(i^*, m^*, \rho^*, \mathsf{ep}^*)$.

First, the simulation of the game provided by the reduction is perfect, and its running time is about that of $\mathcal{A}$. Second, note that if Equation 2 holds, then $\mathcal{B}_2$ breaks the multi-user target collision resistance of IncEnc. We get:

$$\left|\mathsf{Adv}^{\mathsf{Game.1}}_{\mathsf{SIG},N}(\mathcal{A}) - \mathsf{Adv}^{\mathsf{Game.2}}_{\mathsf{SIG},N}(\mathcal{A})\right| \le \mathsf{Adv}^{\mathsf{MULTI\text{-}T\text{-}COLL\text{-}RES},K}_{\mathsf{IncEnc},N,q_s}(\mathcal{B}_2).$$

Let us summarize what we have now, using the same notation as above: if $\mathsf{Game.2}$ outputs 1, then $(m^*, (\rho^*, \sigma^*_{\mathsf{OTS}}, \mathsf{path}^*_{\mathsf{ep}^*})) \neq (m, (\rho, \sigma_{\mathsf{OTS}}, \mathsf{path}_{\mathsf{ep}^*}))$ for user i^*. Due to the changes we have introduced, this means one of the following must hold:

1. $(m, \rho) \neq (m^*, \rho^*)$ and $x \neq x^*$ for $x = \mathsf{IncEnc}(P_{i^*}, m, \rho, \mathsf{ep}^*)$ and $x^* = \mathsf{IncEnc}(P_{i^*}, m^*, \rho^*, \mathsf{ep}^*)$, or
2. $(m, \rho) = (m^*, \rho^*)$ (consequently: $x = x^*$), but $\sigma_{\mathsf{OTS}} \neq \sigma^*_{\mathsf{OTS}}$.

In both cases, we have $(\mathsf{pk}_{i^*,\mathsf{ep}^*}, \mathsf{path}_{i^*,\mathsf{ep}^*}) = (\mathsf{pk}^*_{i^*,\mathsf{ep}^*}, \mathsf{path}^*_{\mathsf{ep}^*})$ with the notation of $\mathsf{Game.1}$. In the following, we will first eliminate the second case, and then focus on the first one.

Game.3: As already mentioned, we now deal with the second case. Namely, we define $\mathsf{Game.3}$ to be exactly as $\mathsf{Game.2}$, but it outputs 0 if $\mathsf{Game.2}$ would output 1 and we are in the second case, i.e., $(m, \rho) = (m^*, \rho^*)$, but $\sigma_{\mathsf{OTS}} \neq \sigma^*_{\mathsf{OTS}}$ and $(\mathsf{pk}_{i^*,\mathsf{ep}^*}, \mathsf{path}_{i^*,\mathsf{ep}^*}) = (\mathsf{pk}^*_{i^*,\mathsf{ep}^*}, \mathsf{path}^*_{\mathsf{ep}^*})$. To bound the difference between $\mathsf{Game.2}$ and $\mathsf{Game.3}$, denote $\sigma_{\mathsf{OTS}} = (y_1, \ldots, y_v)$ and $\sigma^*_{\mathsf{OTS}} = (y^*_1, \ldots, y^*_v)$ and assume we are in this second case. Then, there must be at least one chain $i \in [v]$ such that $y_i \neq y^*_i$. At the same time, because $\mathsf{pk}_{i^*,\mathsf{ep}^*} = \mathsf{pk}^*_{i^*,\mathsf{ep}^*}$, we have

$$\begin{aligned}\mathsf{Chain}_{\mathsf{Th},i,\mathsf{ep}^*}(P_{i^*}, x_i, 2^w - 1 - x_i, y_i) = \mathsf{pk}_{i^*,\mathsf{ep}^*,i} = \mathsf{pk}^*_{i^*,\mathsf{ep}^*,i} &= \mathsf{Chain}_{\mathsf{Th},i,\mathsf{ep}^*}(P_{i^*}, x^*_i, 2^w - 1 - x^*_i, y^*_i) \\ &= \mathsf{Chain}_{\mathsf{Th},i,\mathsf{ep}^*}(P_{i^*}, x_i, 2^w - 1 - x_i, y_i),\end{aligned}$$

where we have used that $x = x^*$. This constitutes a collision somewhere in the chain for user i^*, on the way from y_i (resp. y^*_i) to $\mathsf{pk}_{i^*,\mathsf{ep}^*,i} = \mathsf{pk}^*_{i^*,\mathsf{ep}^*,i}$. More formally, we can build a reduction $\mathcal{B}_3$ that breaks multi-user multi-target collision resistance of Th if we are in this case.

The reduction $\mathcal{B}_3$ works similarly to $\mathcal{B}_1$: it uses the multi-user multi-target collision resistance game with N users and $p = L \cdot v \cdot 2^w$ targets per user. For each user $i \in [N]$, the reduction queries $\mathcal{O}_{\mathsf{target}}(i, \cdot, \cdot)$ to set up all chain elements. After the reveal phase, it simulates $\mathsf{Game.2}$ to $\mathcal{A}$. If the adversary succeeds and we are in the second case, there is a collision in the chain for user i^*, which the reduction can find and use to break the game. We get

$$\left|\mathsf{Adv}^{\mathsf{Game.2}}_{\mathsf{SIG},N}(\mathcal{A}) - \mathsf{Adv}^{\mathsf{Game.3}}_{\mathsf{SIG},N}(\mathcal{A})\right| \le \mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}TCR}}_{\mathsf{Th},N,L\cdot v\cdot 2^w}(\mathcal{B}_3) \le \mathsf{Adv}^{\mathsf{MULTI\text{-}SM\text{-}TCR}}_{\mathsf{Th},N,2\cdot L\cdot v\cdot 2^w}(\mathcal{B}_3)$$

Now that we have ruled out the second case, we can focus on the first case, i.e., the case in which $x \neq x^*$. Note that the verification algorithm checks that $x^* \in \mathcal{C}$ and $x \in \mathcal{C}$ by construction. Therefore, the definition of the incomparable encoding scheme ensures that there exists some $j \in [v]$ such that $x^*_j < x_j$. From now on, j^* denotes the minimum such index.

Game.4 (Final Reduction to Chain Preimage Resistance): We now complete the proof by reducing to multi-user multi-target chain preimage resistance. Recall that if $\mathsf{Game.3}$ outputs 1, we have $x \neq x^*$ and there exists $j^* = \min\{j \in [v] : x^*_j < x_j\}$.

We construct a reduction $\mathcal{B}_4$ against $\mathsf{MULTI\text{-}SM\text{-}CHAIN}_{\mathsf{Th},N,w,L\cdot v}$ as follows:

1. **Query phase:** For each user $i \in [N]$, epoch $\mathsf{ep} \in [L]$, and chain index $j \in [v]$, query $\mathcal{O}_{\mathsf{chain}}(i, T_1^{j,\mathsf{ep}}, \ldots, T_{2^w-1}^{j,\mathsf{ep}})$ to receive the final chain value $\mathsf{pk}_{i,\mathsf{ep},j}$. Store the mapping from (i, ep, j) to the corresponding chain index in Chains_i. This uses $L \cdot v$ chain queries per user.

2. **After reveal phase:** Receive $(P_1, \ldots, P_N)$ from the game. For each user $i \in [N]$, compute the Merkle tree from the one-time public keys $\mathsf{pk}_{i,\mathsf{ep}} = (\mathsf{pk}_{i,\mathsf{ep},1}, \ldots, \mathsf{pk}_{i,\mathsf{ep},v})$ for $\mathsf{ep} \in [L]$ to obtain root_i. Set $\mathsf{pk}_i := (\mathsf{root}_i, P_i)$ and provide $(\mathsf{pk}_1, \ldots, \mathsf{pk}_N)$ to $\mathcal{A}$.

3. **Simulating $\mathcal{O}_{\mathsf{Sig}}(i, \mathsf{ep}, m)$:**

   (a) Compute the encoding $x = \mathsf{IncEnc}(P_i, m, \rho, \mathsf{ep})$ by sampling ρ appropriately (if after K attempts, no ρ is found, return $\perp$).

   (b) For each $j \in [v]$: query $\mathcal{O}_{\mathsf{reveal}}(i, \text{chain index for } (i, \mathsf{ep}, j), x_j)$ to obtain y_j, the chain value at position x_j.

   (c) Compute the Merkle path $\mathsf{path}_{\mathsf{ep}}$ and return $\sigma := (\rho, (y_1, \ldots, y_v), \mathsf{path}_{\mathsf{ep}})$.

4. **Extracting the forgery:** When $\mathcal{A}$ outputs $(i^*, \mathsf{ep}^*, m^*, \sigma^*)$ with $\sigma^* = (\rho^*, (y_1^*, \ldots, y_v^*), \mathsf{path}^*_{\mathsf{ep}^*})$:

   (a) Compute $x^* = \mathsf{IncEnc}(P_{i^*}, m^*, \rho^*, \mathsf{ep}^*)$ and find $j^* = \min\{j : x_j^* < x_j\}$.

   (b) Compute $z' := \mathsf{Chain}_{\mathsf{Th},j^*,\mathsf{ep}^*}(P_{i^*}, x_{j^*}^*, x_{j^*} - x_{j^*}^* - 1, y_{j^*}^*)$.

   (c) Output $(i^*, \text{chain index for } (i^*, \mathsf{ep}^*, j^*), z')$.

The simulation is perfect, $T(\mathcal{B}_4) \approx T(\mathcal{A})$, and $\mathcal{B}_4$ wins the $\mathsf{MULTI\text{-}SM\text{-}CHAIN}$ game whenever $\mathsf{Game.3}$ outputs 1. Therefore:

$$\mathsf{Adv}_{\mathsf{SIG},N}^{\mathsf{Game.3}}(\mathcal{A}) \leq \mathsf{Adv}_{\mathsf{Th},N,w,L\cdot v}^{\mathsf{MULTI\text{-}SM\text{-}CHAIN}}(\mathcal{B}_4).$$

Conclusion: Summing over all game hops, we obtain the desired result. □

A Annex: Technical Results

A.1 Birthday Bound

Lemma A.1 (Birthday Bound)**.** *If k values are sampled independently and uniformly at random from a set of size n, the probability that at least two values are equal is at most $k^2/(2n)$.*

Proof. Let $X_1, \ldots, X_k$ be the sampled values. By the union bound over all $\binom{k}{2}$ pairs:

$$\Pr[\exists i \neq j : X_i = X_j] \leq \sum_{1 \leq i < j \leq k} \Pr[X_i = X_j] = \binom{k}{2} \cdot \frac{1}{n} = \frac{k(k-1)}{2n} \leq \frac{k^2}{2n}.$$ □

References

[1] J. Drake, D. Khovratovich, M. Kudinov, and B. Wagner, “Hash-based multi-signatures for post-quantum ethereum,” Cryptology ePrint Archive, Paper 2025/055, 2025. [Online]. Available: https://eprint.iacr.org/2025/055